

Drought and aquatic community resilience: the role of eggs and seeds in sediments of temporary wetlands

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SUMMARY

1. A long-lived bank of propagules consisting of eggs, seeds and spores is one mechanism that allows aquatic communities to survive drought. A drying (drought) event is, for aquatic organisms in a temporary wetland, a phase from which communities must recover. Such a dry phase is often considered a disturbance but should not be considered adverse or catastrophic for the organisms that have evolved to live in temporarily wet habitats.
2. This paper explores the parallels between the egg bank of zooplankton and the seed bank of aquatic plants as means of survival in temporary wetlands. The resilience of communities in temporary wetland ecosystems is assessed by examining dormancy, hatching, germination, establishment and reproduction of animals and plants from the egg and seed banks of wetlands with a range of wetting and drying regimes.
3. Both the zooplankton and aquatic plants of the temporary wetlands studied rely on their egg and seed banks as a means for surviving drying. These communities recover after the disturbance of drying by means of specific patterns of dormancy, dormancy breakage, hatching, germination, establishment and reproduction. Spatial and temporal patterns of species richness allow resilience through dormancy, as not all species are present at all sites and not all species hatch and germinate at the same time. Multiple generations in the egg and seed bank and complexity of environmental cues for dormancy breakage also contribute to the ecosystem's ability to recover after a drying event. A persistent egg and seed bank allows species-rich communities to hatch, germinate and develop rapidly once dormancy is broken. Rapid establishment of species-rich communities that reproduce rapidly and leave many propagules in the egg and seed bank also facilitates community recovery on flooding of a temporary wetland after a drying event.
4. To maintain the diversity of temporary wetland communities through droughts and floods we need to manage the dry and wet phases of wetlands. To conserve a wide range of wetland types, we need to maintain a variety of hydrological patterns across the landscape.

Keywords: aquatic plants, egg bank, seed bank, zooplankton

Introduction

Temporary wetlands are common but vary substantially in their frequency, duration and timing of flooding (Boulton & Brock, 1999). Over decades even many 'permanent' wetlands occasionally dry. However, in general, as both the predictability and duration of flooding decrease, and the variability of their water regimes increase, wetlands will have a greater ratio of dry time to wet time (Boulton & Brock, 1999). Temporary wetlands range from those with the most predictable patterns (permanent and semi-permanent), to irregularly filling (intermittent) wetlands, to the least predictable and most temporary (episodic) wetlands (Table 1; Boulton & Brock, 1999). Many once-temporary wetlands located in areas of human settlement have been managed to alter water regimes to make each wetland either more predictably flooded through damming or more predictably dry by draining (Brock, Smith & Jarman, 1999; Brock & Jarman, 2000).

A drying (drought) event is, for aquatic organisms in a temporary wetland, a phase from which aquatic communities must recover. Such a dry phase is often considered a disturbance (Bonis, 1998; Boulton, 2003; Lake, 2003; Magoulick & Kobza, 2003) but should not be considered adverse or catastrophic for the organisms that have evolved to live in temporarily wet habitats (Nielsen *et al.*, 2002).

Each species, and therefore the community, is resilient as they survive the dry phase to recover

activity after a disturbance, when the wetland re-floods. How quickly the community returns to an equilibrium following a perturbation is often used as a measure of resilience (Pimm, 1984). The ability of a community to resist change (resistance) and to recover after a disturbance (resilience) together represent ecological stability (MacGillivray & Grime, 1995; Johnson *et al.*, 1996), and some indicators of resistance such as persistence of resting stages can lead to resilience.

For temporary wetlands, resilience can be taken to mean the ability to return to the wet phase community structure and function after a drying event (Brock, 1998). Organisms can persist as resting stages or disperse and recolonise (McPeck & Kalisz, 1998). Biota with poor dispersal mechanisms must persist in the presence of a perturbation such as drying and re-establish when water returns (Begon, Harper & Townsend, 1996). The production of dormant propagules is a way of persisting in the wetland in the face of unpredictable environments (Hairston, 1998) that occurs in many zooplankton, phytoplankton and aquatic plants. Alternatively, it may be a mechanism that enables these biota to exploit the predator- and competitor-free environment after reflooding (Williams, 1998).

Animals and plants survive both the dry and flooded phases of a temporary wetland because they have physiological, morphological, behavioural or life-cycle traits that allow survival in both phases (Wiggins, MacKay & Smith, 1982). Dormant aquatic

Drying pattern	Predictability and duration of filling
Semi-permanent or near permanent	Usually holds some water Annual inflows are greater than minimum loss in 90% of years
Seasonal	May dry during extreme drought events Alternately wet and dry every year according to season Fills during wet season and dries annually Surface water persists for months
Intermittent	Alternately wet and dry but less frequently and regularly than seasonal wetlands Surface water may persist for months to years
Episodic	Annual inflow is less than minimum loss in 90% of years Dry for most of the time Only rarely and irregularly flooded when water may persist for months
Ephemeral	Only fills for a few days after unpredictable rainfall and runoff

Table 1 Classification of water regimes of temporary wetlands (modified from Boulton & Brock, 1999)

plant seeds and vegetative propagules and zooplankton eggs exist in substantial numbers in temporary wetland sediments (Leck, 1989; Finlayson, Cowie & Bailey, 1990; Brock, Theodore & O'Donnell, 1994; Brock, 1998; Brock & Rogers, 1998; Nielsen *et al.*, 2000). The seed and egg banks of temporary wetlands have been shown to be long lived (Boulton & Lloyd, 1992; Jenkins & Briggs, 1997; Brock, 1998; Leck & Brock, 2000). They provide not only refuges from drying but also a means of maintaining genetic, phenotypic, species and community diversity (Ellner & Hairston, 1994; Hairston, Kearns & Ellner, 1996; Leck & Brock, 2000).

This paper aims to: (a) explore the parallels between the egg bank of zooplankton and the seed bank of aquatic plants as means of survival in temporary wetlands, and (b) assess temporary wetland resilience by examining dormancy, hatching and germination, and establishment of communities from the egg and seed banks of wetlands with a range of wetting and drying regimes.

Methods

Criteria for assessing resilience

Those factors that give a community the ability to recover after a disturbance can be used as indicators of resilience. For this paper, we have selected a subset of these indicators that are relevant to temporary wetland communities (Brock, 1998). Fig. 1 outlines the life-cycle stages of organisms in temporary wetlands, indicating factors that could be assessed as indicators of resilience. Many of these are related to the life-cycle stage that persists in the dry phase – the egg and seed bank. Support for the selection of these indicators of resilience comes from either the literature or data presented in this paper (Fig. 1).

Zooplankton and aquatic plants have been chosen for comparison as both groups have dormant eggs or seeds that play a role in the maintenance of temporary wetlands through wet and dry phases (Boulton & Lloyd, 1992; Brock & Casanova, 1997; Bonis, 1998; Brock, 1998; Brock & Rogers, 1998; Jenkins & Boulton, 1998; Nielsen *et al.*, 2000).

The community of dormant propagules in the sediment may contain diploid eggs or ephippia of zooplankton, haploid or diploid spores of phytoplankton, and diploid seeds or asexual propagules of

aquatic plants. In this paper, we refer to this dormant community as the 'egg and seed bank'. We used hatching of zooplankton or germination of aquatic plant seeds, from this dormant covert community, as the indicator of an ability to generate a new active overt community.

Data for assessing resilience

We have chosen data sets from zooplankton hatching studies in three temporary wetlands along the mid-reaches of the River Murray near Albury, NSW, Australia (36°09'S, 146°56'E, 150 m asl): a semi-permanent floodplain wetland (Ryan's 1 billabong), an intermittent floodplain wetland (Ryan's 3 billabong) and an episodic wetland (Ryan's floodplain). A comparable data set was chosen from aquatic plant germination studies in three temporary upland wetlands on the New England Tablelands of NSW (31°S, 151°E, 1000–1325 m asl): a semi-permanent wetland (Llangothlin Lagoon), an intermittent wetland (Racecourse Lagoon) and a seasonal wetland (Barleyfields Lagoon) (Table 2). These wetlands represent a variety of types of temporary wetlands in each regional landscape.

Published plant species numbers from an experimental trial in 1991 (Figs 2, 3 and 5) were used for comparison of aquatic plants (Brock, 1998); seed longevity data up to 1998 (Fig. 3) were from Leck & Brock (2000). Previously unpublished data for aquatic plants are species numbers from germination and field trials between 1992 and 1995 (Figs 2, 3 and 5) and seed longevity data from 1998 to 2000 (Fig. 4). Previously unpublished data for zooplankton are species numbers recorded from Ryan's 1 between 1975 and 1995 and Ryan's 3 between 1990 and 2000 (Fig. 5) and egg longevity data from mesocosm trials (Figs 2–5).

The experimental zooplankton data were collected from a mesocosm trial examining zooplankton hatching from River Murray floodplain sediments. This experiment was designed to assess the contributions to rotifer diversity of recolonisation from the egg bank and recolonisation via aerial dispersal. It was carried out in March–April 1999, the Australian late summer–autumn. Dry sediments from the three wetlands were placed into outdoor mesocosms (1.6 × 1.2 m) and filled with filtered rainwater to a depth of 200 mm. Zooplankton samples were

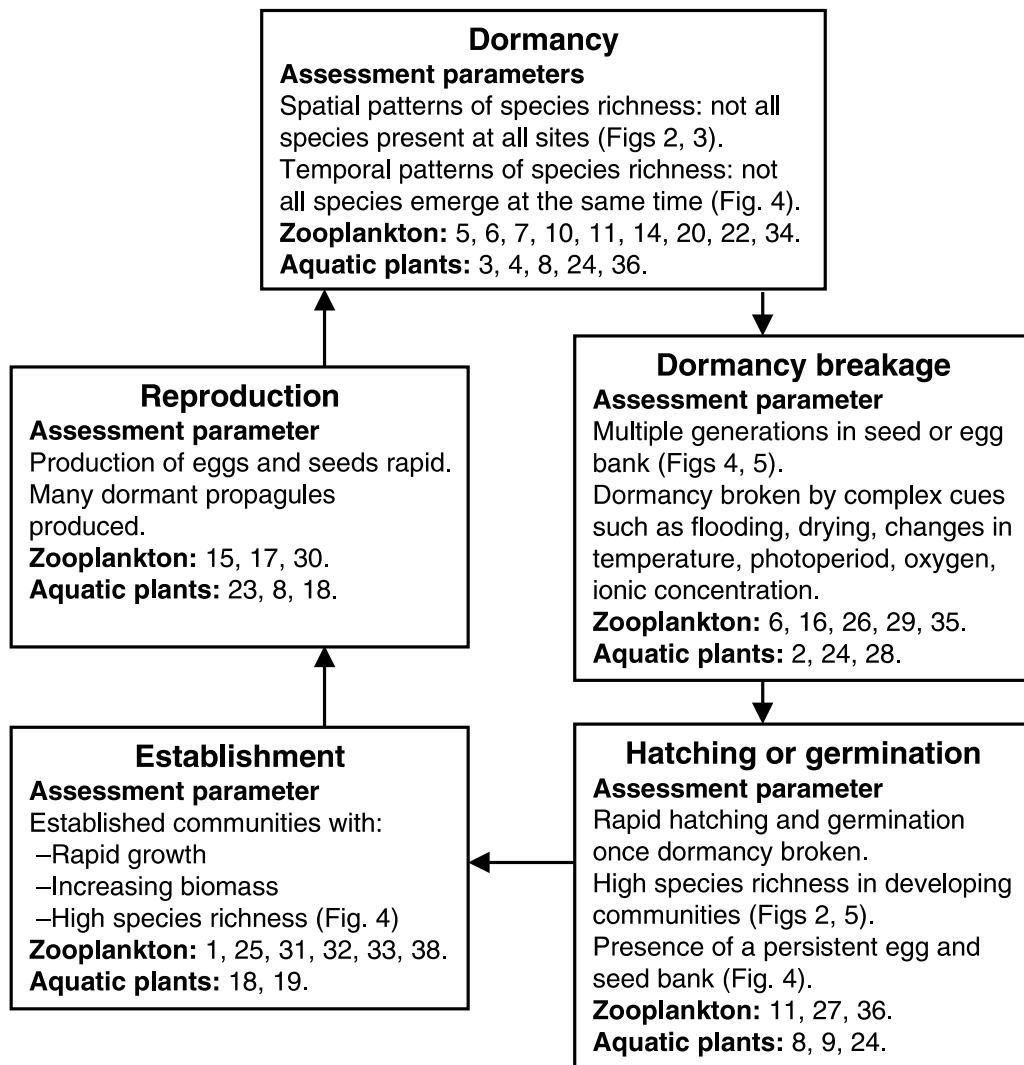


Fig. 1 Assessment parameters for life cycle characteristics that may enhance the ability of communities within temporary wetlands to survive drying events. References: 1. Allen (1976); 2. Baskin & Baskin (1998); 3. Bonis (1998); 4. Bonis *et al.* (1995); 5. Boulton & Lloyd (1992); 6. Brendonck *et al.* (1998); 7. Brendonck & Williams (2000); 8. Brock (1998); 9. Brock & Rogers (1998); 10. De Stasio (1989); 11. De Stasio (1990); 12. Ellner (1996); 13. Ellner & Hairston (1994); 14. Ellner *et al.* (1998); 15. Galimov & Yampolsky (1998); 16. Gilbert (1974); 17. Gilbert (1998); 18. Grillas & Battedou (1998); 19. Grillas *et al.* (1993); 20. Hairston (1996); 21. Hairston, Van Brunt & Kearns (1995); 22. Langley *et al.* (2001); 23. Leck (1989); 24. Leck & Brock (2000); 25. Maier (1992); 26. Medland & Taylor (2001); 27. Nielsen *et al.* (2000); 28. Philippi (1998); 29. Pourriot & Snell (1983); 30. Ricci (2001); 31. Ruttner-Kolisko (1963); 32. Santer (1998); 33. Shiel (1990); 34. Shiel *et al.* (1998); 35. Tan & Shiel (1993); 36. Taylor *et al.* (1990); 37. Thompson, Bakker & Bekker (1997); 38. Wyngaard, Taylor & Mahoney (1991).

collected using a modified Haney–Patalas perspex sampler (4 L). Six 4-L samples were collected and combined from each mesocosm on each sampling occasion. The samples were filtered through a 35- μ m mesh net and preserved in 70% ethanol. The mesocosms were sampled for 30 days after flooding and the abundance and diversity of animals recorded; Langley *et al.* (2001) give a full description of methods. The sediments used had a known history of

inundation and drying (Table 2) and sites were within 0.5 km of each other. In 1994, a major flood had inundated all three wetlands for an extended period. At the time of collection, the semi-permanent wetland was dry for the first time in over 100 years (J. Ryan, personal communication).

The aquatic plant data were collected from glass-house germination experiments in which trays of sediment cores, collected from each wetland in 1991,

Table 2 Temporary wetland sites for seed and egg bank collections used in the present paper: zooplankton (River Murray floodplain) and aquatic plants (New England upland wetlands)

Wetland drying pattern	Zooplankton River Murray floodplain	Aquatic plants Northern Tableland Upland Wetlands
Semi-permanent	Ryan's 1	Llangothlin Lagoon
Seasonal		Barleyfields Lagoon
Intermittent	Ryan's 3	Racecourse Lagoon
Episodic	Floodplain between Ryan's 1 and Ryan's 3	

1992 and 1993, were dried and the propagules let germinate under flooded, waterlogged and fluctuating water regime treatments (Brock *et al.*, 1994). Seedling presence and abundance were recorded by species. The species growing at sites at the time of sediment collection were recorded and monitored along permanent transects each season from 1991 to 1995.

The number of zooplankton species in each of the egg banks was estimated by counting the number of species that hatched during the 30-day experiment. Data from Ryan's 1 (1975–1995) and Ryan's 3 (1990–2000) provided the number of species previously recorded from field samples taken in each wetland.

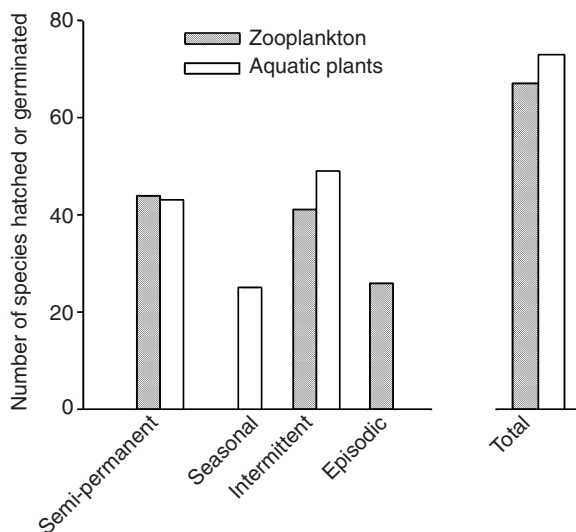


Fig. 2 Number of zooplankton and aquatic plant species hatching and germinating from the four wetland types differing in water regimes. Data on plant species numbers are from Brock (1998) in addition to original data.

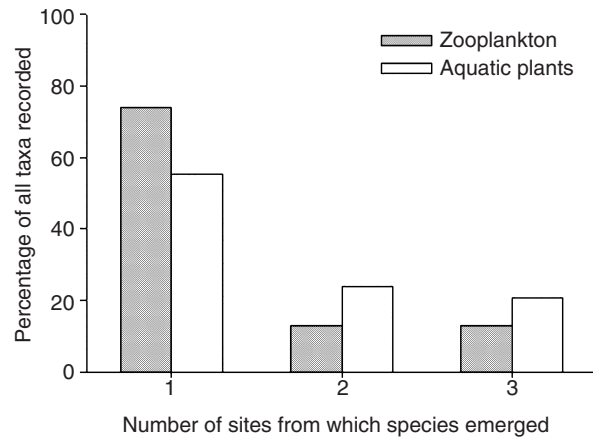


Fig. 3 Uniqueness of zooplankton and aquatic plant communities as measured by the number of wetlands from which a given zooplankton or aquatic plant species hatched or germinated. Data on plant species numbers and seed longevity are from Brock (1998) and Leck & Brock (2000) in addition to original data.

Total species richness of the growing (overt) vegetation from 1991 to 1995 was calculated from species presence scored from surveys of permanent transects in each season at each site.

The longevity of the zooplankton egg bank was assessed by comparing the number of species that

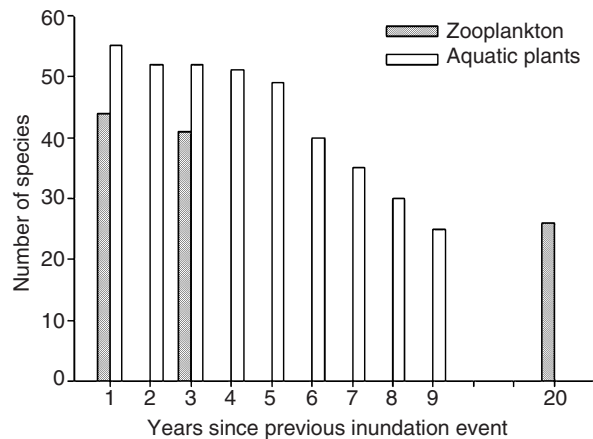


Fig. 4 Longevity of the zooplankton egg bank and aquatic plant seed bank in wetlands considered in the present study as measured by the number of species hatching or germinating upon flooding after a number of years since inundation. Zooplankton longevity was assessed by comparing the number of species that hatched from the episodic, intermittent and semi-permanent wetlands after 1, 3 and 20 years since inundation. Aquatic plant longevity was assessed by comparing the number of species that germinated from sub-samples taken from the semi-permanent, seasonal and intermittent wetlands after 1–9 years since inundation.

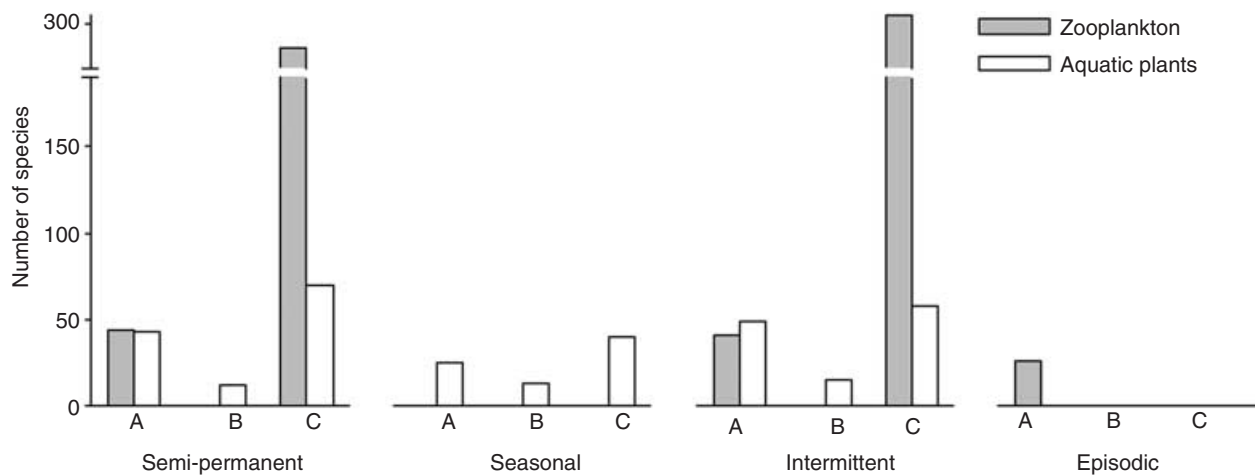


Fig. 5 The potential reservoir of zooplankton and aquatic plant species contained within the egg and seed bank for each wetland in this study. Data presented for comparisons are: Column A, the number of species recorded during zooplankton hatching trials (30 days) and aquatic plant germination trials (100 days); Column B, total number of species recorded at the time of sediment collection for the trial (aquatic plants only); Column C, total number of species recorded growing in the wetland in the previous 25 years (zooplankton) and for the 5 years from 1991 to 1995 (aquatic plants).

hatched from sediments that had been dry for approximately 20 years (episodic wetland), 3 years (intermittent wetland) and for <1 year (semi-permanent wetland). We assumed that at some point in history the egg bank of all three wetlands was homogeneous and that no processes apart from length of time dry (or wet) influenced the number of viable eggs.

The longevity of the aquatic plant seed banks was assessed for semi-permanent, seasonal and intermittent wetlands, from seed bank sediment collected from each wetland site in 1991, dried and stored indoors in insect-proof bins. Each year in spring from 1991 to 2000, five replicate sub-samples per wetland site were placed to germinate under waterlogged conditions in a glasshouse. Species presence and abundance were recorded after 8 weeks. Data from Leck & Brock (2000) for 1991–1997 were combined with new data to 2000.

The uniqueness of zooplankton and aquatic plant communities reflected in the egg and seed bank was examined by plotting the percentage of the total number of zooplankton or aquatic plant species germinating or hatching in one, two or all of the sites.

Results

All wetlands had species-rich seed and egg banks of both zooplankton and aquatic plants with a total of

67 zooplankton and 73 aquatic plant species hatching and germinating. Species number per wetland ranged from 25 aquatic plant species in the seasonal wetland and 26 zooplankton species in the episodic wetland to 44 zooplankton and 43 aquatic plant species in the semi permanent wetlands to 41 zooplankton and 49 aquatic plant species in the intermittent wetlands (Fig. 2).

The compositions of emerging communities were unique to individual wetlands with the majority of species germinating or hatching from only one wetland (Fig. 3). Whereas 74% of the zooplankton and 55% of the aquatic plant species occurred in one site only (Fig. 3), only 26% of the zooplankton species and 45% of the aquatic plants were recorded from more than one of the wetland sites sampled.

Both zooplankton and aquatic plants had persistent seed and egg banks surviving in dry sediments for many years (Fig. 4). After 7-year dry storage, 64% of the original 55 aquatic plant seed bank species germinated from the seed bank. After 9 years, 45% of these species were still germinating (Fig. 4). A zooplankton community comprising 44, 41 and 26 species hatched from sediments collected from sites that had been dry for 1, 3 and 20 years respectively (Fig. 4). The long-lived egg and seed bank (Fig. 4) is not exhausted by a single wetting event or for aquatic plants by successive wetting events (Brock, 1998; Leck & Brock, 2000; Shiel, Green & Tan, 2002).

Communities that become established within newly flooded wetlands are species rich (Figs 3 and 5). Establishment is rapid with up to 44 zooplankton species hatching in each wetland within 30 days of flooding and 49 aquatic plant species germinating in each wetland within 100 days of flooding (Fig. 5, Columns A). Within these experiments, many of the species completed their life cycle to replenish the egg and seed bank (Brock, 1998; Leck & Brock, 2000).

The number of aquatic plant species that germinated from the seed bank in each wetland after flooding (Fig. 5, Columns A) was two to four times greater than the number of aquatic plant species recorded at the time of sediment collection (Fig. 5, Columns B). Records of zooplankton in the field at the time of sediment collection were not available for comparison; however, fewer species of both zooplankton and aquatic plants hatched or germinated (Fig. 5, Columns A) than the total number of species recorded at each wetland where long term records were available (Fig. 5, Columns C; zooplankton 20 years, aquatic plants 5 years): approximately 30% of the total recorded zooplankton species hatched and 65–85% of the total recorded aquatic plant species germinated in the experimental trials (Fig. 5, Columns A & C). Thus, the number of species remaining in the dormant, covert community appears to be 0.5–5 times greater than in the active overt community, indicating that not all eggs hatch or seeds germinate at any given time (Fig. 5).

Discussion

In spite of their differences in phylogeny and organism size, the zooplankton egg bank and aquatic plant seed bank exhibit similar characteristics (Fig. 1) that allow resilience (Figs 2–5).

Dormancy

The species-rich seed and egg banks of both zooplankton and aquatic plants in all wetlands studied (Fig. 2) reflect the natural spatial species-richness of zooplankton and plants in wetlands with a range of wetting and drying patterns. The active communities were unique to the individual wetland with the majority of species hatching and germinating from only one wetland (Fig. 3). Both zooplankton and aquatic plants have persistent egg and seed banks

that are long-lived and survive in dry sediments for many years (Fig. 4).

These findings confirm that both zooplankton and aquatic plants of temporary wetlands can deposit large reservoirs of dormant eggs (De Stasio, 1989; Brendonck & Williams, 2000; Nielsen *et al.*, 2000) and seeds (Leck, 1989; Bonis, Lepart & Grillas, 1995; Brock, 1998). These eggs and seeds can persist for extended periods without a wetting event (Gilbert, 1974; Boulton & Lloyd, 1992; Jenkins & Briggs, 1997; Brock, 1998; Leck & Brock, 2000), although with some loss of viability over time (De Stasio, 1989; Boulton & Lloyd, 1992; Brock, 1998; Leck & Brock, 2000; Langley *et al.*, 2001). For aquatic plant species, this high degree of persistence in temporary wetlands that have unpredictable wetting and drying events contrasts with the transient seed banks of wetlands that have more predictable wetting and drying patterns such as freshwater tidal marshes (Baskin & Baskin, 1998; Leck & Brock, 2000).

Dormancy breakage

The egg and seed banks of the wetlands studied are long-lived, can survive long dry periods and are not exhausted by a single wetting event or successive wetting events (Fig. 4; De Stasio, 1989; Brock, 1998; Leck & Brock, 2000). If the seed and egg bank is not exhausted by the first wetting event, multiple generations can accumulate within the seed and egg bank. The persistence of dormant individuals through wetting episodes suggests that the cues for breaking dormancy are complex and diverse. This allows different communities to develop across a large spatial scale (Fig. 3) and ensures that a substantial component of the species pool will remain dormant as a potential reservoir within the seed and egg bank (Fig. 5).

Evidence for resilience is provided by this persistence of the seed and egg banks as dormancy breakage produces an active overt generation without exhausting the seed bank. This ability to persist results in overlapping generations and this maintains both phenotypic and genetic variability within populations (Gilbert, 1974; Templeton & Levin, 1979; De Stasio, 1990; Ellner, 1996; Bonis, 1998; Ellner, Hairston & Babai, 1998). The occurrence of overlapping generations in some planktonic zooplankton (e.g. rotifers) is partly due to differential hatching caused by

morphological differences in the shells of dormant eggs produced by individuals within populations (Gilbert, 1974, 1995). For aquatic plants, it is because of differential dormancy of seeds caused by factors such as position in the buried seed bank (Baskin & Baskin, 1998). Dormancy breakage cues are not well understood for many wetland plants (Baskin & Baskin, 1998).

The multi-generation seed and egg banks, which are not exhausted by a single wetting event, are maintained by the complexity and diversity of the cues required for initiation of germination and hatching (Taylor, Wyngaard & Mahoney, 1990; Baskin & Baskin, 1998). For example, in North American prairie pothole wetlands, the environmental conditions present at a particular time select the community from the seed and egg bank. When saline conditions are present, salinity-tolerant species will hatch and germinate, whereas in freshwater conditions a community with freshwater-tolerant species may establish (Halse, Shiel & Williams, 1998; Euliss, Wrubleski & Mushet, 1999; Skinner, Sheldon & Walker, 2001).

Hatching and germination from the egg and seed bank

The presence of a large non-active covert community (seed and egg bank) within wetlands, which hatches or germinates rapidly following wetting, also provides resilience (Figs 1 and 5). This has been demonstrated for temporary wetlands on many continents (Leck, 1989; De Stasio, 1990; Grillas *et al.*, 1993; Bonis *et al.*, 1995; Brock, 1998; Ellner *et al.*, 1998). In most wetlands, the active or overt community is less species-rich at any given time than the community that is dormant in the seed and egg banks. The covert community is species-rich for both animals and plants. For example, 270 zooplankton species have been identified from Ryan's 1 billabong but only 20–30 are usually recorded on any given sampling occasion (Boon *et al.*, 1990). The irregular germination and hatching from the seed and egg bank makes it possible for the genotypes of populations to change during a season. This provides the potential for a community with a different range of species and genetic compositions to colonise at any given time (Gilbert, 1974; De Stasio, 1990; Ellner & Hairston, 1994; Bonis, 1998; Callaghan, 1998; Grillas & Battedou, 1998; Ricci, 2001).

Persistence of the seed and egg banks protects the wetland community from wetting events that are not long enough in duration to allow successful hatching or germination and replenishment of the seed and egg bank. This longevity of seeds and eggs partially explains the species richness of communities and the dissimilarity of community composition among closely neighbouring wetlands (Fig. 3, De Stasio, 1990; Shiel, Green & Nielsen, 1998; Brock *et al.*, 1999; Brock & Jarman, 2000).

Establishment and reproduction

For both zooplankton and aquatic plants, establishment of overt populations and communities from the egg and seed bank is rapid (Fig. 5). Many temporary wetland plant species progress from germination to seed dispersal in 8–12 weeks, which is rapid for angiosperms (Brock, 1998; Brock & Rogers, 1998). Zooplankton are even faster. Zooplankton hatching from dry sediment in some cases has been recorded as beginning within days of wetting (Nielsen *et al.*, 2000) and continues as additional species take a progressively longer time to respond to the same triggers or respond to a new set of triggers as the environment changes (Makarewicz & Likens, 1975). Typically, rotifers live only for a few days but it has been estimated that about 10^{12} individuals could develop from a single egg in 60 days (Ruttner-Kolisko, 1963). Within a single season, populations usually have multiple generations asexually and sexually produced resting eggs (Gilbert, 1980). What causes the change from asexual to sexual reproduction and the production of resting eggs is not fully understood, but it could be initiated by stimuli such as high population density, temperature, photoperiod and starvation (Gilbert, 1995, 1998; Ricci, 2001).

Resilience

Both the zooplankton and aquatic plants of the wetlands studied rely on their egg and seed banks as a means for surviving drying in temporary wetlands. These communities recover after the disturbance of drying by means of their specific patterns of dormancy, dormancy breakage, hatching, germination, establishment and reproduction (Fig. 1). The data presented in Figs 2–5 support the assessment parameters defined for resilience of each life-cycle

stage in Fig. 1. Spatial and temporal patterns of species richness allow resilience through dormancy, because not all species are present at all sites (Figs 2 and 3) and not all species hatch and germinate at the same time (Fig. 4). Multiple generations in the egg and seed bank (Figs 4 and 5) and complexity of environmental cues for dormancy breakage also contribute to the community's ability to recover after a drying event. A persistent egg and seed bank allows individuals in species-rich communities to hatch, germinate and develop rapidly once dormancy is broken (Figs 2, 4 and 5). Rapid establishment of species-rich communities which reproduce rapidly and leave many propagules in the egg and seed bank also facilitates community recovery on flooding of a temporary wetland after a drying event (Fig. 4).

Many zooplankton and aquatic plant species persist within the dormant propagule bank and for them drying is not a catastrophic event. Many of these species may also disperse between wetlands by wind or animal transport and thus can colonise other sites. For other groups, such as insects, fish, frogs, turtles, birds and mammals, that also utilise these unpredictable temporary wetlands, drying can result in catastrophic population declines, as in general they must disperse and recolonise as drying and then re-wetting occurs (Wiggins *et al.*, 1982; Williams, 1985; Williams, 1996). This may be less true of some vertebrate species that aestivate on site in seasonal temporary wetlands, for example the endangered Western Swamp Tortoise, *Pseudemydura umbrina*, in Australia (Burbidge, 1981, 1991).

This complex of survival mechanisms not only allows survival but also provides heterogeneity of biotic communities in space and time at a range of scales. This is exemplified by our zooplankton and aquatic plant communities that are species rich and heterogeneously distributed among wetlands at any one time, and within wetlands over time.

Management implications and recommendations

Manipulation of the water regime of wetlands is common and has ecological consequences. The egg and seed bank represents a substantial component of wetland diversity (Brendonck & Williams, 2000). Modification of the water regime will select for different components of wetland plant communities based on their response to the presence or absence of

water at germination, establishment and reproductive life-cycle phases and thus seed bank replenishment is modified (Brock & Casanova, 1997; Casanova & Brock, 2000; Leck & Brock, 2000). Previous zooplankton studies have also indicated that modification of the water regime may modify replenishment of the egg bank (Nielsen *et al.*, 2000). More permanent flooding (or the loss of drying) removes one of the cues necessary to trigger hatching of some zooplankton species (Neckles, Murkin & Cooper, 1990; Golladay, Taylor & Palik, 1997; Nielsen *et al.*, 2000). Long-term loss of flooding appears to result in loss of viability of eggs (Boulton & Lloyd, 1992; Langley *et al.*, 2001). A loss of drying may also result in the loss of biodiversity within these systems (Nielsen *et al.*, 2000). This suggests that drying is an important process within some wetland ecosystems.

Water regime modification has made many wetlands more permanently flooded and other wetlands more permanently dry, reducing the heterogeneity of wetlands in the landscape (Brock *et al.*, 1999; Brock & Jarman, 2000). The effect of such alterations to water regime on the survival and composition of seed and egg banks will have important implications for subsequent communities.

To maintain the diversity of temporary wetland communities through 'droughts and flooding rains' (Dorothea McKellar's poem 'My Country'), we need to manage the dry and wet phases of wetlands. We also need to conserve a wide range of wetland types by providing and maintaining a variety of hydrological patterns across the landscape.

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References

- Allen J.D. (1976) Life history patterns in zooplankton. *The American Naturalist*, **110**, 165–180.
- Baskin C.C. & Baskin J.M. (1998) *Seeds: Ecology, Biogeography and Evolution of Dormancy and Germination*. Academic Press, San Diego, CA, USA, 666 pp.
- Begon M., Harper J.L. & Townsend C.R. (1996) *Ecology: Individuals, Populations and Communities*. Blackwell Science, London, UK, 945 pp.
- Bonis A. (1998) Coexistence in a variable and unpredictable habitat: the importance of the seed bank and contrasted regeneration niche for annual plants in temporary pools. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 53–69.
- Bonis A., Lepart J. & Grillas P. (1995) Seed bank dynamics and coexistence of annual macrophytes in a temporary and variable habitat. *Oikos*, **74**, 81–92.
- Boon P.I., Frankenberg J., Hillman T.J., Oliver R.L. & Shiel R.J. (1990) Billabongs. In: *The Murray* (Eds N. Mackay & D. Eastburn), pp. 183–198. CSIRO Publications Canberra, Australia.
- Boulton A.J. (2003) Parallels and contrasts in the effects of drought on stream macroinvertebrate assemblages. *Freshwater Biology*, **48**, 1173–1185.
- Boulton A.J. & Brock M.A. (1999) *Australian Freshwater Ecology. Processes and Management*. Gleneagles Publishing, Adelaide, Australia, 250 pp.
- Boulton A.J. & Lloyd L.N. (1992) Flooding frequency and invertebrate emergence from dry floodplain sediments of the River Murray, Australia. *Regulated Rivers: Research and Management*, **7**, 137–151.
- Brendonck L. & Williams W.D. (2000) Biodiversity in wetlands of dry regions (drylands). In: *Biodiversity in Wetlands: Assessment, Function and Conservation* (Eds B. Gopal, W.J. Junk & J.A. Davis), pp. 181–194. Backhuys Publishers, Leiden, The Netherlands.
- Brendonck L., Riddoch B., Van De Weghe V. & Van Dooren, T. (1998) The maintenance of egg banks in very short-lived pools – a case study with anostracans (Branchiopoda). *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 141–161.
- Brock M.A. (1998) Are temporary wetlands resilient? Evidence from seed banks of Australian and South African wetlands. In: *Wetlands for the Future* (Eds A.J. McComb & J.A. Davis), pp. 193–206. Gleneagles Press, Adelaide, Australia.
- Brock M.A. & Casanova M.T. (1997) Plant life at the edges of wetlands; ecological responses to wetting and drying patterns. In: *Frontiers in Ecology: Building the Links* (Eds N. Klomp & I. Lunt), pp. 181–192. Elsevier Science, Oxford, UK.
- Brock M.A. & Jarman P.J. (2000) Wetland use and conservation in the agricultural environment: managing processes for the components. In: *Nature Conservation 5; Conservation in Production Environments: Managing the Matrix* (Eds J.L. Craig, D.A. Saunders & N. Mitchell), pp. 258–268. Surrey Beatty & Sons, Chipping Norton, Australia.
- Brock M.A. & Rogers K.A. (1998) The regeneration potential of the seed bank of an ephemeral floodplain in South Africa. *Aquatic Botany*, **61**, 123–135.
- Brock M.A., Theodore K.A. & O'Donnell L. (1994) Seed-bank methods for Australian Wetlands. *Australian Journal of Marine and Freshwater Research*, **45**, 483–493.
- Brock M.A., Smith R.G.B. & Jarman P.J. (1999) Drain it, dam it: alteration of water regime in shallow wetlands on the New England Tableland of NSW. *Wetlands Ecology and Management*, **7**, 37–46.
- Burbidge A.A. (1981) The ecology of the Western Swamp Tortoise, *Pseudemydura umbrina* (Testudines, Chelidae). *Australian Wildlife Research*, **8**, 203–222.
- Burbidge A.A. (1991) What the tortoise taught us. *Landscape*, **6**, 28–34.
- Callaghan T.P. (1998) Clonal variation in the degree of asynchrony: multiple germination strategies within a bryozoan population. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 85–96.
- Casanova M.T. & Brock M.A. (2000) How do depth, duration and frequency of flooding influence the establishment of wetland plant communities? *Plant Ecology*, **147**, 237–250.
- De Stasio B.T. Jr (1989) The seedbank of a freshwater crustacean: copepodology for the plant ecologist. *Ecology*, **70**, 1377–1389.
- De Stasio B.T. Jr (1990) The role of dormancy and emergence patterns in the dynamics of a freshwater zooplankton community. *Limnology and Oceanography*, **35**, 1079–1090.
- Ellner S. (1996) Environmental fluctuations and the maintenance of genetic diversity in age or stage-structured populations. *Bulletin of Mathematical Biology*, **58**, 103–127.
- Ellner S. & Hairston N.G. Jr (1994) Role of overlapping generations in maintaining genetic variation in a fluctuating environment. *The American Naturalist*, **143**, 403–417.
- Ellner S.P., Hairston N.G. Jr & Babai D. (1998) Long-term diapause and spreading of the risk across the life cycle. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 297–312.

- Euliss N.H. Jr, Wrubleski D.A. & Mushet D.M. (1999) Wetlands of the Prairie Pothole Region: invertebrate species composition, ecology, and management. In: *Invertebrates in Freshwater Wetlands of North America: Ecology and Management* (Eds D.P. Batzer, R.B. Rader & S.A. Wissinger), pp. 471–514. John Wiley & Sons Inc, New York, USA.
- Finlayson C.M., Cowie I.D. & Bailey B.J. (1990) Sediment seed banks in grasslands in the Magela Creek Floodplain, Northern Australia. *Aquatic Botany*, **38**, 163–176.
- Galimov Y.R. & Yampolsky L.Y. (1998) Optimal strategies of investment in resting stage production and germination in unpredictable seasonal environments. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 313–326.
- Gilbert J.J. (1974) Dormancy in rotifers. *Transactions of the American Microscopical Society*, **93**, 490–513.
- Gilbert J.J. (1980) Female polymorphism and sexual reproduction in the rotifer *Asplanchna*: evolution of their relationship and control by dietary tocopherol. *The American Naturalist*, **116**, 409–431.
- Gilbert J.J. (1995) Structure, development and induction of a new diapause stage in rotifers. *Freshwater Biology*, **34**, 263–270.
- Gilbert J.J. (1998) Asexual diapause in the rotifer *Synchaeta*: diversified bet-hedging, energetic cost and age effects. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 97–107.
- Golladay S.W., Taylor B.W. & Palik B.J. (1997) Invertebrate communities of forested limesink wetlands in southwest Georgia, USA: habitat use and influence of extended inundation. *Wetlands*, **17**, 383–393.
- Grillas P. & Battedou G. (1998) Effects of flooding date on the biomass, species composition and seed production in submerged macrophyte beds in temporary marshes in the Camargue (S. France). In: *Wetlands for the Future* (Eds A.J. McComb & J.A. Davis), pp. 207–218. Glencoe Press, Adelaide, Australia.
- Grillas P., Garcia-Murillo P., Geertz-Hansen O., Marba N., Montes C., Duarte C.M., Tan-Ham L. & Grossman A. (1993) Submerged macrophyte seed bank in a temporary marsh (Doñana National Park, SW Spain): abundance and correlation with established vegetation. *Oecologia*, **94**, 1–6.
- Hairton N.G. Jr (1996) Zooplankton egg banks as biotic reservoirs in changing environments. *Limnology and Oceanography*, **41**, 1087–1092.
- Hairton N.G. Jr (1998) Time travellers: what's timely in diapause research? *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 1–15.
- Hairton, N.G. Jr, Van Brunt, R.A. & Kearns, C.M. (1995) Age and survivorship of diapausing eggs in a sediment egg bank. *Ecology*, **76**, 1706–1711.
- Hairton, N.G. Jr, Kearns C.M. & Ellner S.P. (1996) Phenotypic variation in a zooplankton egg bank. *Ecology*, **77**, 2382–2392.
- Halse S.A., Shiel R.J. & Williams W.D. (1998) Aquatic invertebrates of Lake Gregory, northwestern Australia, in relation to salinity and ionic composition. *Hydrobiologia*, **381**, 15–29.
- Jenkins K.M. & Boulton A.J. (1998) Community dynamics of invertebrates emerging from reflooded lake sediments: flood pulse and aeolian influences. *International Journal of Ecology and Environmental Sciences*, **24**, 179–192.
- Jenkins K.M. & Briggs S. (1997) *Wetland Invertebrates and Flood Frequency on Lakes Along Teryaweynya Creek*. NSW National Parks and Wildlife Service, Canberra, Australia.
- Johnson K.H., Vogt K.A., Clark H.J., Schmidt O.J. & Vogt D.J. (1996) Biodiversity and productivity and the stability of ecosystems. *Trends in Ecology and Evolution*, **11**, 372–377.
- Lake P.S. (2003) Ecological effects of perturbation by drought in flowing waters. *Freshwater Biology*, **48**, 1161–1172.
- Langley J.M., Shiel R.J., Nielsen D.L. & Green J.D. (2001) Hatching from the sediment egg-bank, or aerially-dispersed? – The use of mesocosms in assessing rotifer biodiversity. *Hydrobiologia*, **446/447**, 203–211.
- Leck M.A. (1989) Wetland seed banks. In: *Ecology of Soil Seed Banks* (Eds M.A. Leck, V.T. Parker & R.L. Simpson), pp. 283–308. Academic Press, San Diego, CA, USA.
- Leck M.A. & Brock M.A. (2000) Ecological and evolutionary trends in wetlands: evidence from seeds and seed banks in New South Wales, Australia and New Jersey, USA. *Plant Species Biology*, **15**, 97–112.
- MacGillivray C.W. & Grime J.P. (1995) Testing predictions of the resistance and resilience of vegetation subjected to extreme events. *Functional Ecology*, **9**, 640–649.
- McPeck M.A. & Kalisz S. (1998) On the joint evolution of dispersal and dormancy in metapopulations. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 31–51.
- Magoulick D.D. & Kobza R.M. (2003) The role of refugia for fishes during drought: a review and synthesis. *Freshwater Biology*, **48**, 1186–1198.
- Maier G. (1992) *Metacyclops minutus* (Claus, 1863) – population dynamics and life history characteristics of a rapidly developing copepod. *Internationale Revue der gesamten Hydrobiologie*, **77**, 455–466.
- Makarewicz J. & Likens G.E. (1975) Niche analysis of a zooplankton community. *Science*, **190**, 1000–1003.
- Medland V.L. & Taylor B.E. (2001) The strategies of emergence from diapause for cyclopoid copepods in a temporary pond. *Archiv für Hydrobiologie*, **150**, 329–349.

- Neckles H.A., Murkin H.R. & Cooper J.A. (1990) Influences of seasonal flooding on macroinvertebrate abundance in wetland habitats. *Freshwater Biology*, **23**, 311–322.
- Nielsen D.L., Smith F.J., Hillman T.J. & Shiel R.J. (2000) Impact of water regime and fish predation on zooplankton resting egg production and emergence. *Journal of Plankton Research*, **22**, 433–446.
- Nielsen D.L., Hillman T.J., Smith F.J. & Shiel R.J. (2002) The influence of seasonality and duration of flooding on zooplankton in experimental billabongs. *River Research and Applications*, **18**, 227–237.
- Philippi T.E. (1998) Prolonged diapause and models of species coexistence: a cautionary tale from annual plants in deserts. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 19–31.
- Pimm S.L. (1984) The complexity and stability of ecosystems. *Nature*, **307**, 321–326.
- Pourriot R. & Snell T.W. (1983) Resting eggs in rotifers. *Hydrobiologia*, **104**, 213–224.
- Ricci C. (2001) Dormancy patterns in rotifers. *Hydrobiologia*, **446/447**, 1–11.
- Ruttner-Kolisko A. (1963) The relationships of the Rotatoria. In: *The Lower Metazoa* (Ed. E.C. Dougherty), pp. 263–272. University of California Press, Berkeley and Los Angeles, USA.
- Santer B. (1998) Life cycle strategies of free-living copepods in freshwaters. *Journal of Marine Systems*, **15**, 327–336.
- Shiel R.J. (1990) Zooplankton. In: *The Murray* (Eds N. Mackay & D. Eastburn), pp. 275–284. CSIRO Publications, Canberra, Australia.
- Shiel R.J., Green J.D. & Nielsen D.L. (1998) Floodplain biodiversity: why are there so many species? *Hydrobiologia*, **387/388**, 39–46.
- Shiel R.J., Green J.D. & Tan L.W. (2002) Microfaunal and resting-stage heterogeneity in ephemeral floodplain pools, upper River Murray floodplain, Australia. *Verhandlungen der Internationalen Vereinigung für Theoretische und Angewandte Limnologie*, **27**, 3738–3741.
- Skinner R., Sheldon F. & Walker K.F. (2001) Animal propagules in dry wetland sediments as indicators of ecological health: effects of salinity. *Regulated Rivers: Research and Management*, **17**, 191–197.
- Tan L. & Shiel R.J. (1993) Responses of billabong rotifer communities to inundation. *Hydrobiologia*, **255/256**, 361–369.
- Taylor B.E., Wyngaard G.A. & Mahoney D.L. (1990) Hatching of *Diaptomus stagnalis* eggs from a temporary pond after a prolonged dry period. *Archiv für Hydrobiologie*, **117**, 271–278.
- Templeton A.R. & Levin D.A. (1979) The evolutionary consequences of seed pools. *The American Naturalist*, **114**, 232–249.
- Thompson K., Bakker J. & Bekker R. (1997) *The Soil Seed Banks of North West Europe: Methodology, Density and Longevity*. Cambridge University Press, Cambridge, United Kingdom, 276 pp.
- Wiggins G.B., Mackay R.J. & Smith I.M. (1982) Evolutionary and ecological strategies of animals in annual temporary pools. *Archiv für Hydrobiologie, Supplement*, **58**, 97–206.
- Williams W.D. (1985) Biotic adaptations in temporary lentic waters, with special reference to those in semi-arid and arid regions. *Hydrobiologia*, **125**, 85–110.
- Williams D.D. (1996) Environmental constraints in temporary fresh waters and their consequences for the insect fauna. *Journal of the North American Benthological Society*, **15**, 634–650.
- Williams D.D. (1998) The role of dormancy in the evolution and structure of temporary waters. *Archiv für Hydrobiologie, Advances in Limnology*, **52**, 109–124.
- Wyngaard G.A., Taylor B.E. & Mahoney D.L. (1991) Emergence and dynamics of cyclopoid copepods in an unpredictable environment. *Freshwater Biology*, **25**, 219–232.

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